Aryan Pandey

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Education

University at Buffalo, State University of New York — Buffalo, NY
Master of Science (M.S.) in Computer Science and Engineering, Research Track — *Jan 2025 – Dec 2026 (Expected)*

University at Buffalo, State University of New York — Buffalo, NY
Bachelor of Science (B.S.) in Computer Engineering (GPA: 3.7, Dean's List) — *Aug 2018 – May 2022*

Experience

Research Assistant — Aug 2026 – Present
Empire AI — *Buffalo, NY*

- Helped develop the UB AI Innovation Marketplace, a website concept connecting institutional challenges with responsible AI ideas, technical expertise, and communities of interest.
- Supporting Empire AI's applied AI research agenda by helping identify high-impact problems and shape research directions in housing, childcare, and other domains.
- Contributing to the ideation, prototyping, implementation, and evaluation of AI systems designed for real-world use, with attention to stakeholder needs, responsible deployment, and measurable outcomes.

Research Software Engineering Fellow — Jun 2026 – Aug 2026
Princeton University – Princeton Neuroscience Institute — *Princeton, NJ*

- Built the distributed fitting backend for *PsyNeuLink*, an open-source cognitive modeling library, so researchers can fit models across a SLURM cluster instead of one machine.
- Scaled close to linearly out to **128+ cores** on Dask, launching a worker cluster inside the SLURM job and recovering the same parameters as a single-machine fit.
- Implemented hierarchical fitting with partial pooling: subject parameters are drawn from a group-level distribution, fit by expectation-maximization over KDE likelihoods.
- Added a neural likelihood estimation path with Pyro and SBI, exposed alongside KDE as an option in the fitting API.
- Validated against HSSM, an established Bayesian package, to confirm the distributed version recovers the same parameters.

Software Engineer — Jun 2022 – Apr 2024
Axon — *Remote*

- Cut cloud storage costs by **$1.5M/month**, helping build an archival system on Azure with Temporal and Blob Storage.
- Sped up evidence-video ingestion by roughly **1.5**× on a platform holding **250 PB** of data.
- Held on-call for globally deployed ingestion of body-cam, TASER, and CJIS data, keeping Dockerized services above **99.99%** availability.
- Built Scala microservices for back-end infrastructure, with unit, integration, and load testing across the services I owned.
- Shipped monthly releases to every production region through Kubernetes and Puppet.
- Cut time-to-detect on incidents by adding Grafana dashboards and restructuring how our services logged to Splunk.

Teaching Assistant – VLSI, Computer Architecture, Computer Security — Aug 2025 – Present
University at Buffalo — *Buffalo, NY*

- Ran labs on CMOS logic and standard-cell design: inverter sizing, NAND/NOR cells, ring oscillators, and SRAM bitcells.
- Taught the Cadence Virtuoso flow end to end, from schematic through layout, parasitic extraction, and DRC/LVS sign-off.
- Mentored teams through mixed-signal IC projects (DPLLs, TRNGs) from spec to tapeout, including the timing and power trade-offs behind each block.
- Wrote lab notes, assignments, and exams; ran weekly office hours for combined undergraduate and graduate sections.

Teaching Assistant – Computer Security, Computer Organization — Aug 2020 – Dec 2021
University at Buffalo — *Buffalo, NY*

- Ran weekly recitations on CPU architecture, ALU design, and pipelining; led exam review sessions.
- Covered malware, authentication, OS and network security, and cryptography; demoed common vulnerabilities and their countermeasures.

Software Engineer – Research Internship Jan 2020 – Aug 2020
University at Buffalo *Buffalo, NY*

- Wrote the C back-end for a portable FASTQ/A DNA analysis machine, with an inotify watchdog feeding new files into the pipeline.
- Parallelized the pipeline with OpenMP for an **8**× speedup over the single-threaded version.

Projects

Recurrent Differentiable Weightless Networks (RDWN) | *PyTorch, Verilog, Vivado* Feb 2026 – Present

- Added recurrent connections to Differentiable Weightless Networks, a LUT-only architecture that replaces multiply-accumulates with table lookups mapping onto FPGA LUT-6 primitives.
- Reached **82.2%** test accuracy on Spiking Heidelberg Digits against **67%** for the feedforward baseline, and **97.75%** on N-MNIST.
- Wrote the PyTorch → Verilog export tool for trained truth tables and wiring, then ran the model end to end on FPGA at **0.6W**.

ResearchShelf | *Python, FastAPI, Vue 3, SQLite, Docker, Claude Code* Jul 2026 – Present

- Built and released a self-hosted research-paper platform: team libraries with role-based access, REST APIs, and a browser-based PDF reader.
- Pulls papers from Zotero, DOI/arXiv, Crossref, OpenAlex, and Unpaywall, dedupes by SHA-256, and indexes full text with SQLite FTS5.
- Published on PyPI, with tests running automatically across Linux, macOS, and Windows on Python 3.11 and 3.12.
- LLM agents reach the same authenticated workflows over MCP.

Multi-Objective NAS for Spiking Neural Networks | *PyTorch, snnTorch, REINFORCE* Jan 2026 – May 2026

- Trained an LSTM controller with REINFORCE to search seven architecture dimensions — depth, width, membrane leak, threshold, recurrence, neuron model, and per-layer learning rule — against a joint accuracy, sparsity, and spike-timing objective.
- Gated the sparsity and timing rewards below 1.5× chance accuracy, keeping silent networks from inflating the baseline and washing out the advantage signal.
- Over 128 evaluations on Spiking Heidelberg Digits, the controller concentrated **96%** of its sampling mass on one architecture, up from **7%** at uniform, and beat matched-compute random search on reward.
- Showed the reward's timing weight, not the search itself, decided which architecture won, by decomposing both candidates' rewards into accuracy and bonus terms.

CUDA Optimization for On-the-Fly Training | *CUDA, PyTorch, Triton* Mar 2025 – May 2025

- Targeted on-the-fly training, where synthetic samples are generated on the device but the default path copies every batch back to host and returns it.
- Kept the generated samples resident on the GPU and copied back only on a mismatch, cutting end-to-end training time by roughly **20%**.
- Took a stress-test step from **46.71s to 39.34s** by overlapping transfers with compute using pinned memory and multiple streams.
- Profiled the pipeline to find where the GPU sat idle, then fixed it with stream scheduling and batching changes.

Technical Skills

Languages: Python, C/C++, CUDA, Scala, Java, SQL, JavaScript, Verilog, Bash, ARM/x86/MIPS Assembly
Machine Learning & Scientific Computing: PyTorch, snnTorch, Norse, Pyro/SBI, Hugging Face, Triton, NumPy/SciPy, Dask, OpenMP, Nsight profiling
Backend & Infrastructure: FastAPI, SQLAlchemy, REST APIs, gRPC/Protobuf, MCP, SQLite/FTS5, pytest, Docker, Kubernetes, SLURM, Azure, Linux, Git, GitHub Actions, CI/CD
Hardware/EDA: Xilinx Vivado, Cadence Virtuoso, SPICE, DRC/LVS, parasitic extraction
Working knowledge: Terraform, Kafka, Spark/Databricks, Cassandra, Temporal, Prometheus/Grafana, Splunk